as thick as a human hair and yet they'd be completely invisible to our most sophisticated instruments. Right now, right next to you, right next to me, and right next to everyone else, there could be another spatial dimension—a dimension beyond left/right, back/forth, and up/down, a dimension that's curled up but still large enough to swallow something as thick as this page—that remains beyond our grasp.[38]

Large Extra Dimensions and Large Strings

By trapping three of the four forces, the braneworld scenario significantly relaxes experimental constraints on how big the extra dimensions can be, but the extra dimensions aren't the only thing this approach allows to get bigger. Drawing on insights of Witten, Joe Lykken, Constantin Bachas, and others, Ignatios Antoniadis, together with Arkani-Hamed, Dimopoulos, and Dvali, realized that in the braneworld scenario even unexcited, low-energy strings can be *much* larger than previously thought. In fact, the two scales—the size of extra dimensions and the size of strings—are closely related.

Remember from the previous chapter that the basic size of string is determined by requiring that its graviton vibrational pattern communicate a gravitational force of the observed strength. The weakness of gravity translates into the string's being very short, about the Planck length (10^{-33} centimeters). But this conclusion is highly dependent on the size of the extra dimensions. The reason is that in string/M-theory, the strength of the gravitational force we observe in our three extended dimensions represents an interplay between two factors. One factor is the intrinsic, fundamental strength of the gravitational force. The second factor is the size of the extra dimensions. The larger the extra dimensions, the more gravity can spill into them and the weaker its force will *appear* in the familiar dimensions. Just as larger pipes yield weaker water pressure because they allow water more room to spread out, so larger extra dimensions yield weaker gravity, because they give gravity more room to spread out.

The original calculations that determined the string's length assumed that the extra dimensions were so tiny, on the order of the Planck length,